

MMET 383 Lab

Instructor: Tianle Zhang

Email: tianle@tamu.edu

Place: THOM 115C

Lab assignments needs to be submitted as a ".zip" file to Canvas;

If you finish your lab assignment during the lab, you may leave early;

A personal laptop is encouraged;

Attendance is required;

Office hour: 3-5pm Monday



You can get this slide
and the video tutorial
from this QR code:



ATM

CS.Fall.2026

Home

Announcements

Simple Syllabus

Modules

Grades

Course Evaluations

Ally Report

Panorama

Mediasite Collection

YuJa Video

Poll Everywhere 2.0

Panopto

Google Gemini

Assignments

Discussions

Files

Item Banks

Pages

People

Quizzes

Rubrics

Outcomes

Collaborations

Syllabus

Settings

New2026-383-Labs1-9.pdf

Download New2026-383-Labs1-9.pdf (572 KB) | Alternative formats

Page < 1 > of 17 ZOOM

Dr. Sheng-Jen (“Tony”) Hsieh

Lab #1 MMET383

Visual Basic Programming: Create a User Interface

Objective: Practice Visual Basic programming to develop a user interface using Tool Box.

Task #0: Listen to and take notes during the lab demonstration.

Task #1: Develop a user-interface using the Tool Box.

This interface will include **Radio Buttons, Image Boxes, and Text Boxes.**
Following are the requirements for the interface:

1. Three Radio buttons, Image boxes, and Text boxes are required within a form.
2. When one of the Radio buttons is pressed, its corresponding image and text box will be shown.
3. When Exit is pressed, the program ends.

Figure 1 shows the layout for the user interface.

Task 1

1 Lay Out the Form: 1 Form, 5 Textboxes, 5 Labels, 4 Buttons

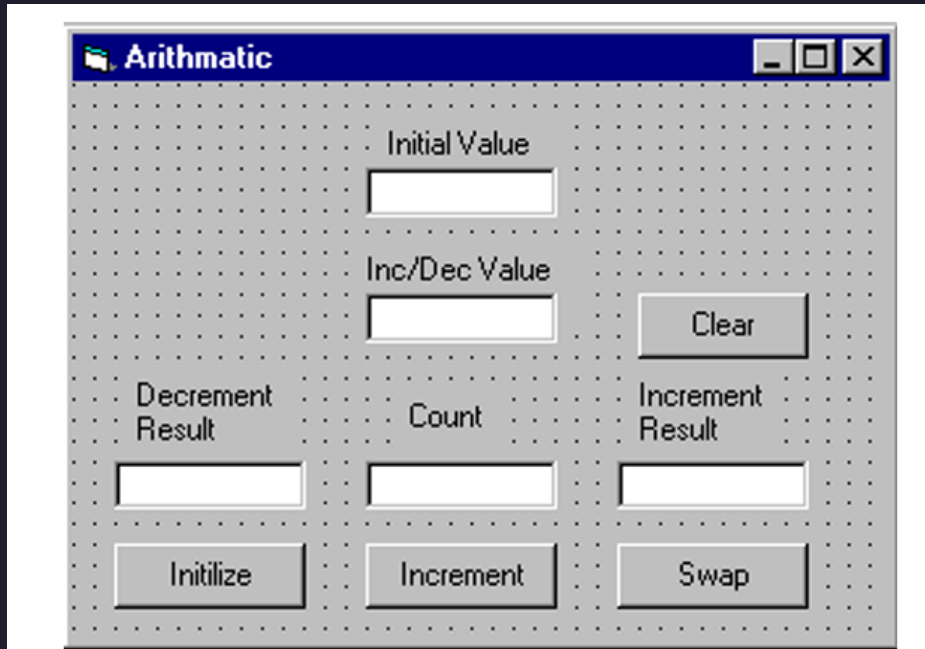


Figure 1. Arithmetic User Interface Layout.

Textboxes · 5

- ▶ `txtInitialValue`
- ▶ `txtIncDecValue`
- ▶ `txtIncrementResult`
- ▶ `txtDecrementResult`
- ▶ `txtCount`

Buttons · 4

- ▶ `btnInitialize`
- ▶ `btnIncrement`
- ▶ `btnDecrement`
- ▶ `btnSwap`

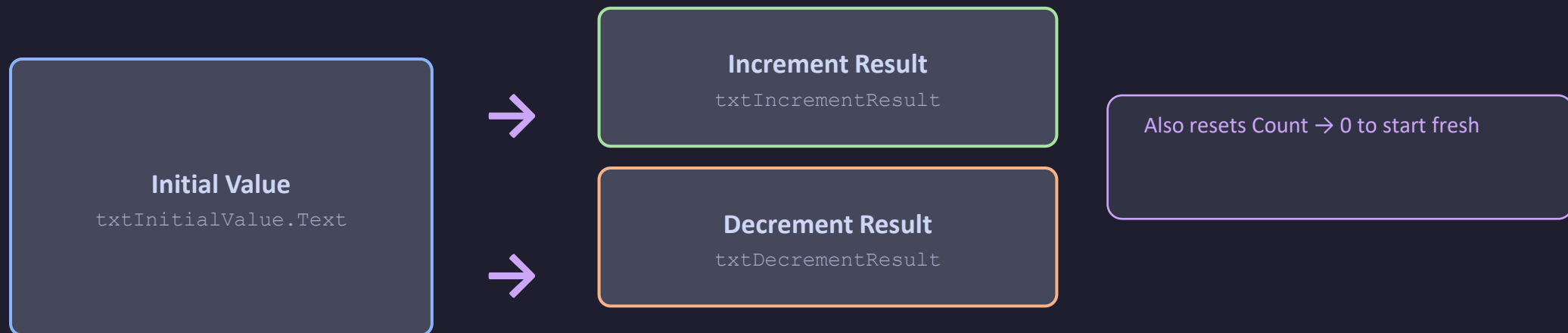
Names must exactly match the control names after Handles in your code

2

Initialize Button: Copy Initial Value into Both Results

Form1.vb - btnInitialize_Click

```
Private Sub btnInitialize_Click(...)
    initialValue = CSng(txtInitialValue.Text)
    incrementResult = initialValue
    decrementResult = initialValue
    txtIncrementResult.Text = CStr(incrementResult)
    txtDecrementResult.Text = CStr(decrementResult)
End Sub
```



3

Increment Button: Calculate First, Then Display

Form1.vb - btnIncrement_Click

```
Private Sub btnIncrement_Click(...)
    incDecValue = CSng(txtIncDecValue.Text)
    ' Calculate first -- only in memory, textbox untouched
    incrementResult = incrementResult + incDecValue
    count = count + 1
    ' Then display -- write back only after calculating
    txtIncrementResult.Text = CStr(incrementResult)
End Sub
```

①

Read Inc/Dec Value

CSng converts to Single



②

Calculate incrementResult

+ incDecValue (calculate first)



③

Increment count

count = count + 1



④

Write back to textbox

CStr displays (then display)

count, incrementResult, and decrementResult are all declared at the top of the Class → values survive between clicks

4

Decrement Button: Symmetric Logic, One Sign Flipped

btnIncrement_Click

```
Private Sub btnIncrement_Click(...)
    incDecValue = CSng(...)

    incrementResult =
        incrementResult + incDecValue

    count = count + 1
    txtIncrementResult.Text =
        CStr(incrementResult)
End Sub
```

▲ Adds with +

btnDecrement_Click

```
Private Sub btnDecrement_Click(...)
    incDecValue = CSng(...)

    decrementResult =
        decrementResult - incDecValue

    count = count + 1
    txtDecrementResult.Text =
        CStr(decrementResult)
End Sub
```

▲ Subtracts with -

5

Count Textbox: Both Buttons Share One Counter

Increment Button



Decrement Button



count

Integer · module-level variable



txtCount.Text

count = count + 1

One click, one increment

count is a module-level variable -- Increment and Decrement write to the same data, so it can track the total number of clicks

6

Swap Button: Swap Two Values with a Temp Variable

Form1.vb - btnSwap_Click

```
temp = incrementResult      ' 1. C = A
incrementResult = decrementResult ' 2. A = B
decrementResult = temp      ' 3. B = C
```

Before Swap

A = "coke"

B = "tea"



After Swap

A = "tea"

B = "coke"

temp is just a stepping stone -- the real goal is swapping A and B's values

7

CStr and CSng: Converting Strings and Numbers

String

Contents of a textbox
`txtXxx.Text`

Read → `CSng( )`

Display ← `CStr( )`

Single

A number you can do math with

Read with `CSng`, display with `CStr`

e.g. `CSng(txtInitialValue.Text)` reads a number / `CStr(incrementResult)` writes it back

Task 2

1

Setup: 1 Form, 5 Buttons, 7 Textbox/Label Pairs

Variables

Investor_Name → **Name**

Investment_Goal → **FV**

Years → **Term**

Growth_Rate → **Rate**

Initial_Investment → **PV**

Number_Of_Investment → **Count**

Total_Initial_Investment → **Total_PV**

Buttons · 5

- ▶ btnCalculate
- ▶ btnClearInput
- ▶ btnClubSummary
- ▶ btnNewInvestor
- ▶ btnExit

Portfolio Summary label + 2 textboxes start hidden — Club Summary reveals them

2

Calculate Button: PV Formula and Running Totals

Form1.vb - btnCalculate_Click

```
Private Sub btnCalculate_Click(...)
    FV = CSng(txtInvestmentGoal.Text)
    Term = CSng(txtTermYears.Text)
    Rate = CSng(txtGrowthRate.Text) / 100
    PV = FV / (1 + Rate) ^ Term
    Total_PV = Total_PV + PV
    Count = Count + 1
    txtInitialInvestment.Text = Format(PV, "$0.00")
End Sub
```

$PV = FV / (1 + Rate) ^ Term$

Total_PV += PV

Count += 1

Every click calculates one investment, then folds it into the running totals

3

Growth Rate: Percent Input, Decimal in the Formula



```
Form1.vb  
  
Rate = CSng(txtGrowthRate.Text) / 100
```

The formula needs a decimal rate, but users naturally type a whole percentage — dividing by 100 bridges the two

4

Currency Formatting: \$ Sign, 2 Decimal Places

Three ways to format the same number

```
FormatCurrency(PV)
Format(PV, "Currency")
Format(PV, "$0.00")
```

```
' all three can produce:  $321.78
```

This project uses `Format(PV, "$0.00")` for Initial Investment and Total Initial Investment

5

Club Summary Button: Reveal the Portfolio Summary

Form1.vb — btnClubSummary_Click

```
Private Sub btnClubSummary_Click(...)
    lblPortfolioSummary.Visible = True
    lblNumberOfInvestments.Visible = True
    txtNumberOfInvestments.Visible = True
    lblTotalInitialInvestment.Visible = True
    txtTotalInitialInvestment.Visible = True
End Sub
```

Visible = False



Visible = True

Same controls, just toggled — hidden by default, revealed on click

6

Clear Input vs. New Investor vs. Exit

Clear Input

Clears only the
current inputs:

Investment Goal
Term in Years
Growth Rate

Totals stay untouched

New Investor

Clears every
textbox, resets
Count and Total_PV
to 0, and hides the
Portfolio Summary
section again

Exit

Ends the program
with a single
statement:

End